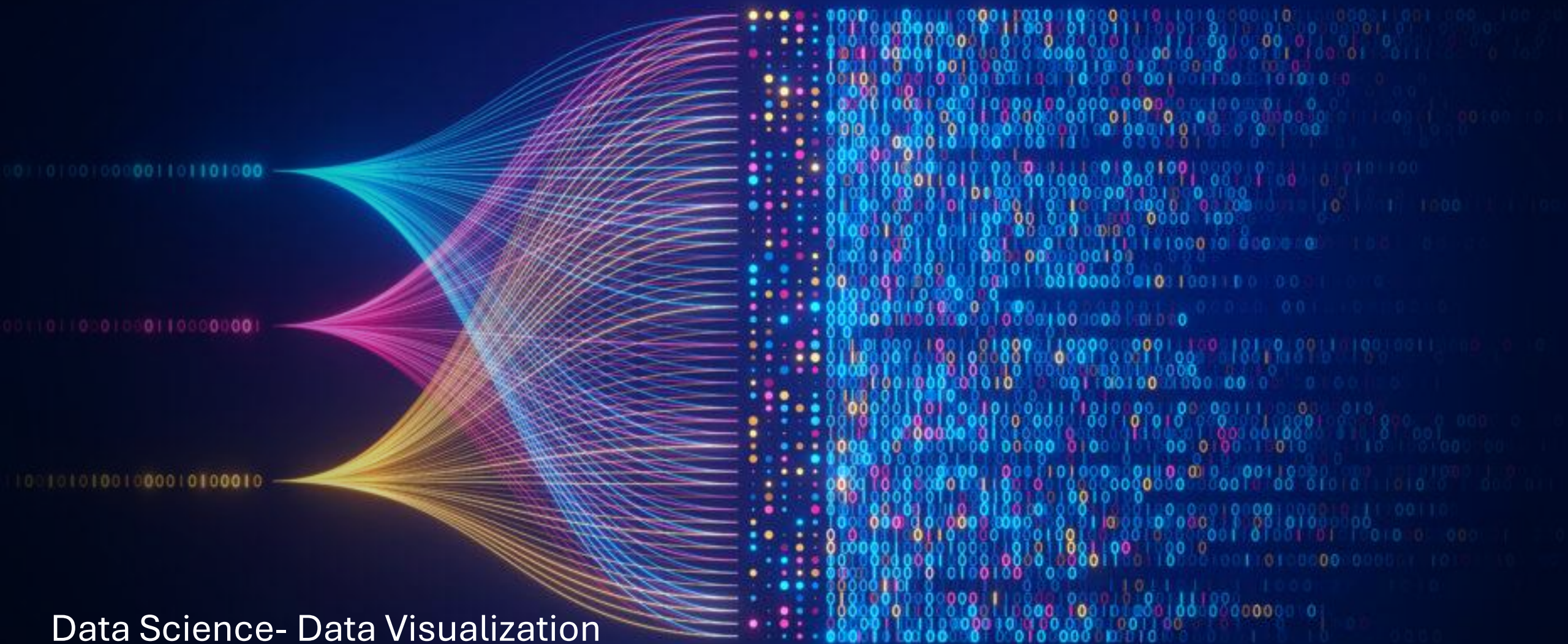




Data Science- Data Visualization

Prof. Alraee

Data- What is it and Why it matters?

Data is information or facts about something.

- It can be numbers, words, pictures, voices, even emotions!

Think of data as clues that help us understand patterns or answer questions.

Example: Collecting data on your classmates' favorite colors helps you see which color is the most popular.

Data is like a puzzle - each piece adds to the bigger picture!

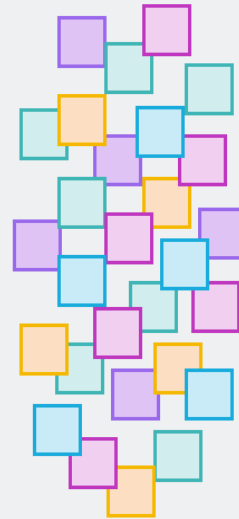
Data Scientists

1- Collect a lot of information (called "data") about things.

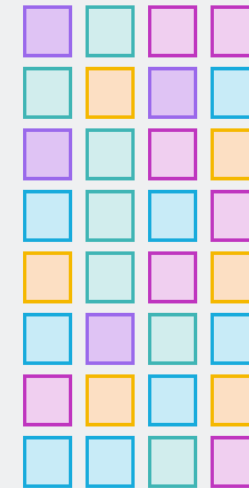
2- Use math and computers to find patterns or solve mysteries in the data.

3- Use the information that they got to make good decisions!

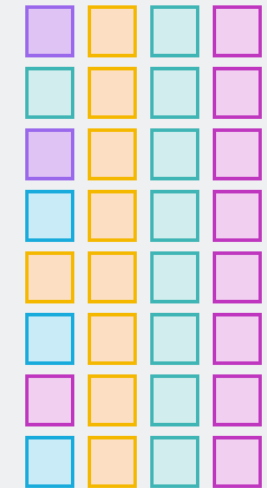
Big Data



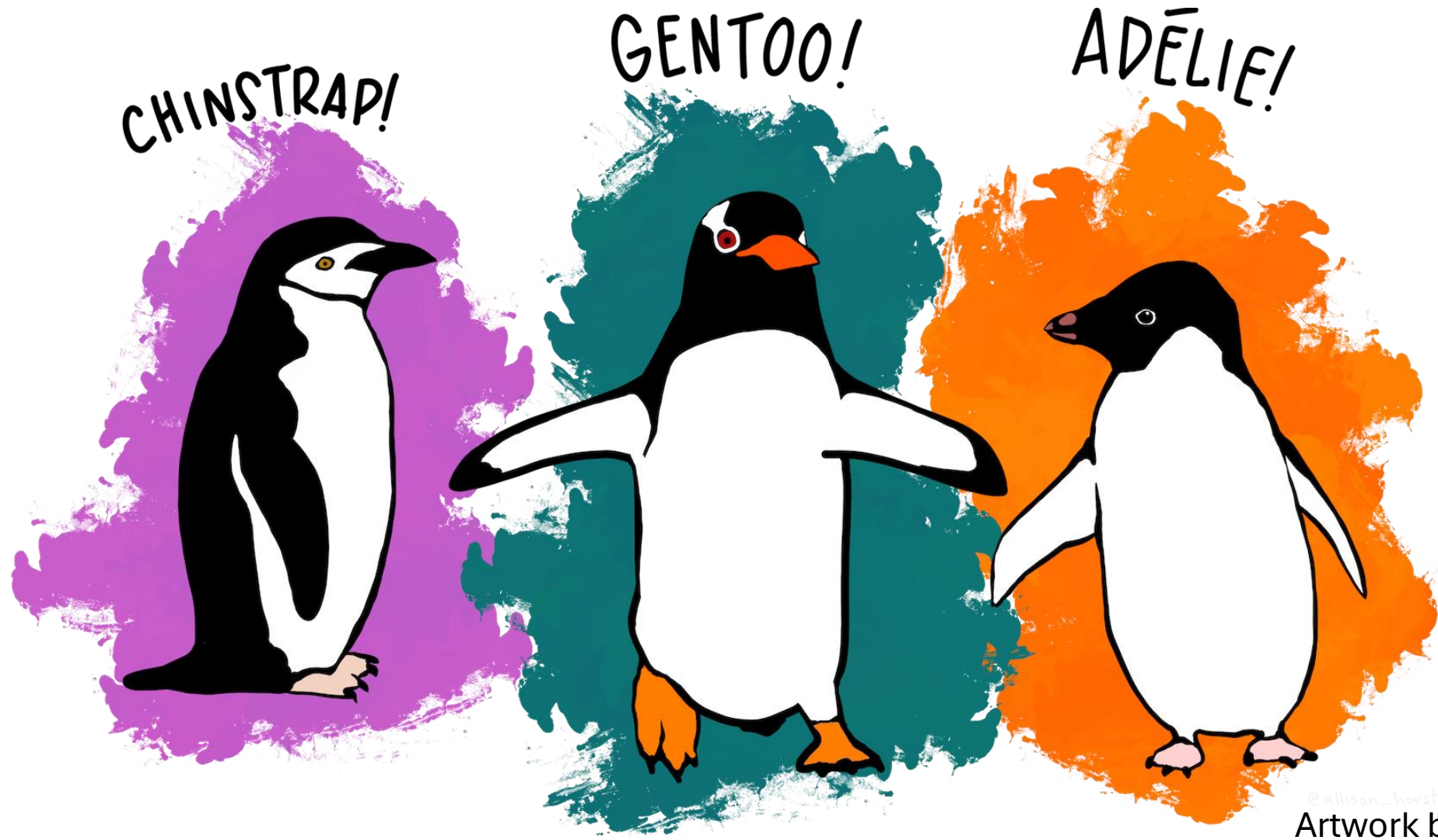
Analytics



Decisions



Penguins Activity



@allison_horst
Artwork by @allison_horst

AutoSave Off

penguins

Search

AM

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelp

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Sensitivity

Add-ins

Analyze Data

Show ToolPak

CommentsShare

A1

species

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex								
2	Adelie	Torgersen	39.1	18.7	181	3750	MALE								
3	Adelie	Torgersen	39.5	17.4	186	3800	FEMALE								
4	Adelie	Torgersen	40.3	18	195	3250	FEMALE								
5	Adelie	Torgersen													
6	Adelie	Torgersen	36.7	19.3	193	3450	FEMALE								
7	Adelie	Torgersen	39.3	20.6	190	3650	MALE								
8	Adelie	Torgersen	38.9	17.8	181	3625	FEMALE								
9	Adelie	Torgersen	39.2	19.6	195	4675	MALE								
10	Adelie	Torgersen	34.1	18.1	193	3475									
11	Adelie	Torgersen	42	20.2	190	4250									
12	Adelie	Torgersen	37.8	17.1	186	3300									
13	Adelie	Torgersen	37.8	17.3	180	3700									
14	Adelie	Torgersen	41.1	17.6	182	3200	FEMALE								
15	Adelie	Torgersen	38.6	21.2	191	3800	MALE								
16	Adelie	Torgersen	34.6	21.1	198	4400	MALE								
17	Adelie	Torgersen	36.6	17.8	185	3700	FEMALE								
18	Adelie	Torgersen	38.7	19	195	3450	FEMALE								
19	Adelie	Torgersen	42.5	20.7	197	4500	MALE								
20	Adelie	Torgersen	34.4	18.4	184	3325	FEMALE								

<>

penguins

+

Data collected

Species:

Gentoo penguins are the fastest underwater swimmers of all penguins, reaching speeds up to 22 miles per hour!

Adélie penguins can walk up to 31 miles to find food and were named after the wife of a French explorer.

Chinstrap penguins get their name from the thin black line under their chin, making it look like they're wearing tiny helmets!

Islands:

Biscoe Island is home to the most diverse penguin population in the dataset, hosting Gentoo, Adélie, and Chinstrap penguins!

Dream Island might sound magical, and it's fitting because it's mostly home to the fun-loving Adélie penguins.

Torgersen Island is small but special, known primarily as a nesting spot for Adélie penguins who thrive there in large colonies.

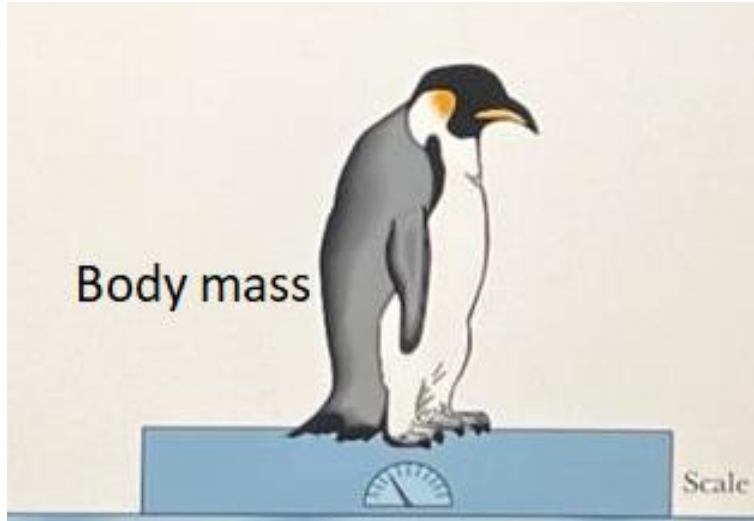
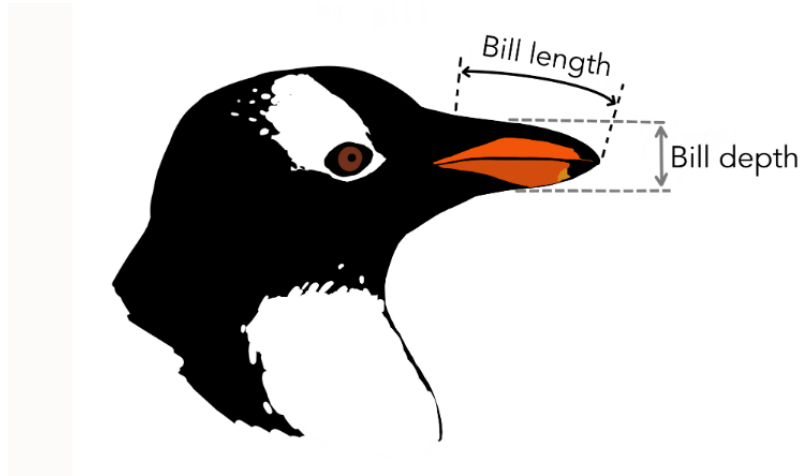


Image created by Twinkl

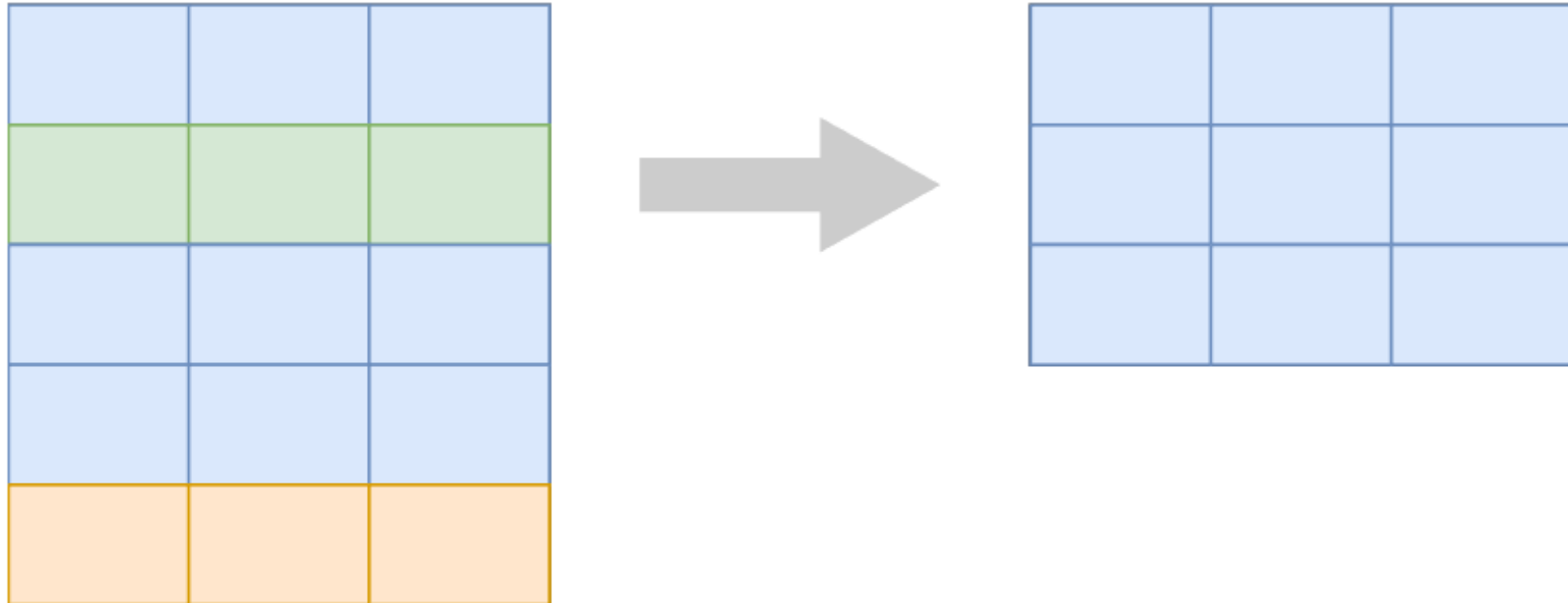


Artwork by @allison_horst

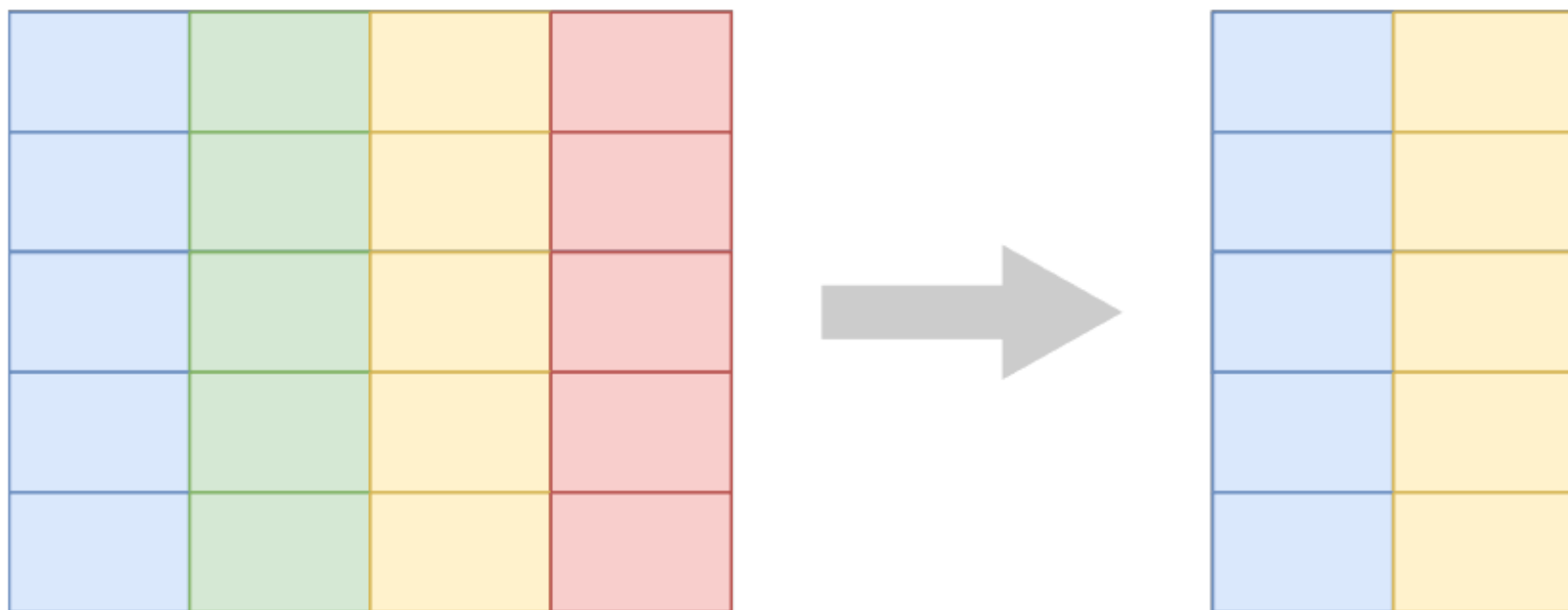


Talk To The Fin!" by Jennifer Hadley

Pick rows from a table: `filter()`



Pick columns from a table: `select()`



Analyze subsets: `group_by()` and `summarize()`

